

SQL Pizza Sales Analysis

Data Analysis Project
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All
About
Pizza!



Tools Used:

- MySQL
- SQL
- Pizza Sales Dataset

Project Overview

Objective:

- Analyze pizza sales data using SQL
- Discover sales trends and customer behaviour
- Generate business insights for better decision-making.

Skills Applied:

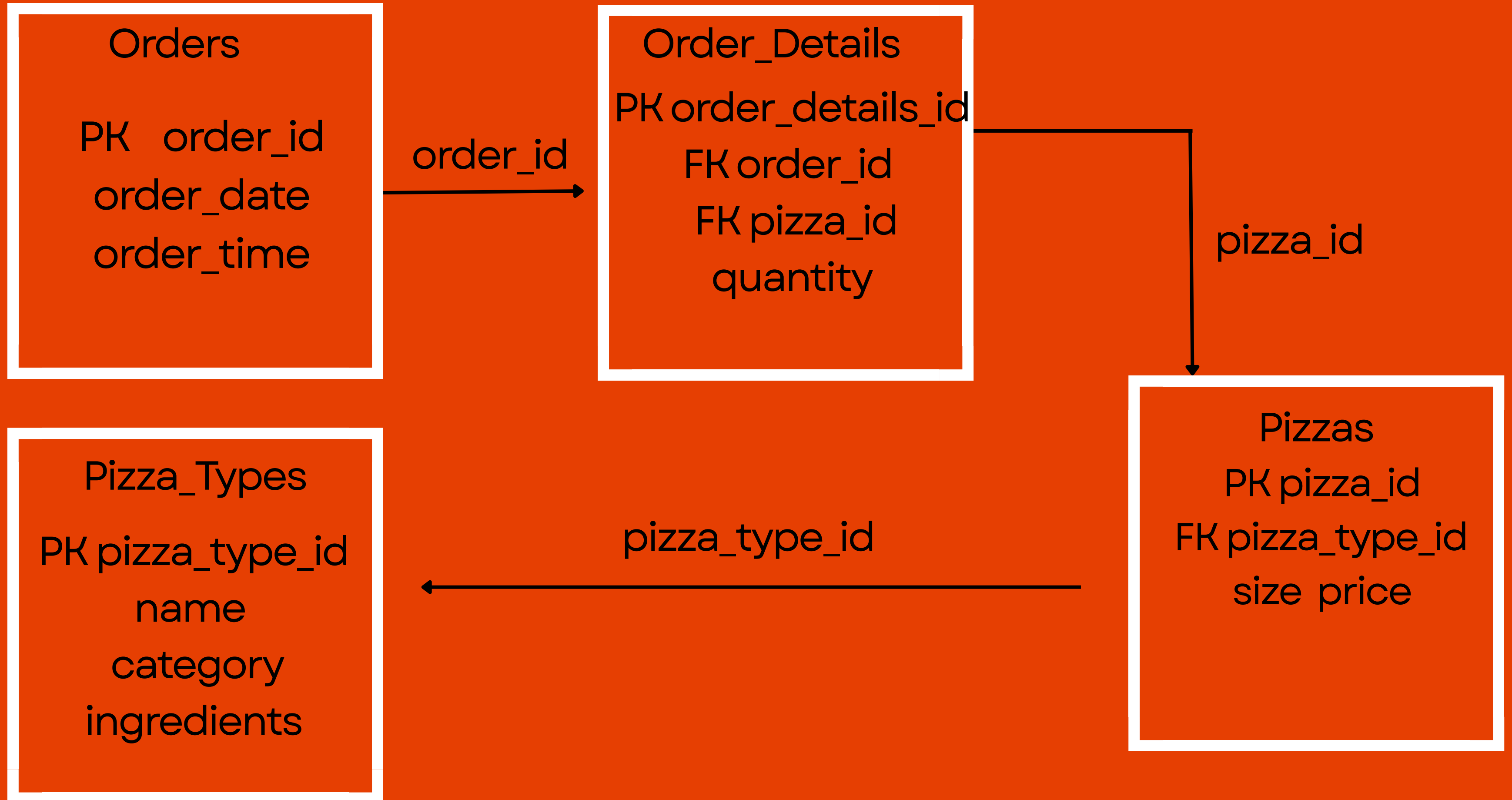
- Joins
- Aggregate Functions
- GROUP BY
- CTEs
- Window Functions
- Data Analysis



Business Questions Solved

1.Retrieve the total number of orders placed.	2.Calculate the total revenue generated from pizza sales.	3.Identify the highest-priced pizza.	4.Identify the most common pizza size ordered
5.List the top five most ordered pizza types along with their quantities.	6.Find the total quantity sold for each pizza category	7.Determine the distribution of orders by hour of the day	8.Analyze the category-wise distribution of pizzas
9.Calculate the average number of pizzas ordered per day	10.Identify the top three pizza types based on revenue.	11.Calculate the percentage contribution of each pizza type to total revenue.	12.Analyze the cumulative revenue generated over time

PIZZA SALES DATABASE SCHEMA



- Retrieve the total number of orders placed.

```
SELECT count(order_id) as total_orders  
FROM orders;
```

Result Grid	
	total_orders
▶	21350

- Calculate the total revenue generated from pizza sales.

```
SELECT  
  round (sum(order_details. quantity * pizzas. price),2) as total_sales  
FROM order_details join Pizzas  
on pizzas. pizza_id = order_details. pizza_id
```

Result Grid	
	total_sales
▶	817860.05

- Identify the highest-price pizza.

```
SELECT
    pizza_types.name, pizzas.price
FROM
    pizza_types
    JOIN
        pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
ORDER BY Pizzas.price DESC
LIMIT 1;
```

Result Grid			Filter Rows:
	name	price	
▶	The Greek Pizza	35.95	

- Identify the most common pizza size ordered.

```
SELECT
    pizzas.size, COUNT(order_details.order_details_id)
FROM
    pizzas
    JOIN
        order_details ON pizzas.pizza_id = order_details.pizza_id
GROUP BY pizzas.size;
```

Result Grid			Filter Rows:
	size	count(order_details.order_details_id)	
▶	M	15385	
	L	18526	
	S	14137	
	XL	544	
	XXL	28	

- List the top 5 most ordered pizza types along with their quantities

```
SELECT
    pizza_types.name, SUM(order_details.quantity) AS quantity
FROM
    pizza_types
    JOIN
    pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
    JOIN
    order_details ON order_details.pizza_id = pizzas.pizza_id
GROUP BY pizza_types.name
ORDER BY quantity DESC
LIMIT 5;
```

	name	quantity
▶	The Classic Deluxe Pizza	2453
	The Barbecue Chicken Pizza	2432
	The Hawaiian Pizza	2422
	The Pepperoni Pizza	2418
	The Thai Chicken Pizza	2371

- Join the necessary tables to find the total quantity of each pizza category ordered

```
SELECT
    pizza_types.category,
    SUM(order_details.quantity) AS quantity
FROM
    pizza_types
    JOIN
    pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
    JOIN
    order_details ON order_details.pizza_id = pizzas.pizza_id
GROUP BY pizza_types.category
ORDER BY quantity DESC;
```

	category	quantity
▶	Classic	14888
	Supreme	11987
	Veggie	11649
	Chicken	11050

- Determine the distribution of orders by hour of the day

```
SELECT
    HOUR(order_time) as Hour, COUNT(order_id) as order_count
FROM
    orders
GROUP BY HOUR(order_time);
```

	Hour	order_count
▶	11	1231
	12	2520
	13	2455
	14	1472
	15	1468
	16	1920
	17	2336
	18	2399
	19	2009
	20	1642
	21	1198
	22	663
	23	28
	10	8
	9	1

- Join relevant tools to find the category the avg number of pizzas ordered per day.

```
SELECT
    category, COUNT(name)
FROM
    pizza_types
GROUP BY category
```

	category	count(name)
▶	Chicken	6
	Classic	8
	Supreme	9
	Veggie	9

- Determine the top 3 most ordered pizzas types based on revenue.

```
SELECT
    pizza_types.name,
    SUM(order_details.quantity * pizzas.price) AS revenue
FROM
    pizza_types
    JOIN
    pizzas ON pizzas.pizza_type_id = pizza_types.pizza_type_id
    JOIN
    order_details ON order_details.pizza_id = pizzas.pizza_id
GROUP BY pizza_types.name
ORDER BY revenue DESC
LIMIT 3;
```

	name	revenue
▶	The Thai Chicken Pizza	43434.25
	The Barbecue Chicken Pizza	42768
	The California Chicken Pizza	41409.5

- Group the orders by date and calculate the average number of pizzas ordered per day

```
SELECT
    AVG(quantity)
FROM
    (SELECT
        orders.order_date, SUM(order_details.quantity) AS quantity
    FROM
        orders
        JOIN order_details ON orders.order_id = order_details.order_id
    GROUP BY orders.order_date) AS order_quantity;
```

	avg (quantity)
▶	138.4749

- Calculate the percentage contribution of each pizza types to total revenue

```
SELECT
    pizza_types.category,
    (SUM(order_details.quantity * pizzas.price) / (SELECT
round (sum(order_details. quantity * pizzas. price),2) as total_sales
FROM order_details join Pizzas
on pizzas. pizza_id = order_details. pizza_id ) ) * 100 as revenue
FROM
    pizza_types
    JOIN
    pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
    JOIN
    order_details ON order_details.pizza_id = pizzas.pizza_id
GROUP BY pizza_types.category
ORDER BY revenue DESC;
```

Result Grid			Filter Rows:
	category	revenue	
▶	Classic	26.90596025566967	
	Supreme	25.45631126009862	
	Chicken	23.955137556847287	
	Veggie	23.682590927384577	

- **Conclusion**
- **Successfully analyzed pizza sales data using SQL.**
- **Extracted key business insights from sales, orders, and revenue data.**
- **Identified top-selling pizzas, customer ordering patterns, and peak sales hours.**
- **Performed data analysis using Joins, Aggregate Functions, CTEs, and Window Functions.**
- **The insights generated from this analysis can help improve pricing strategies, inventory management, and overall business decision-making.**

THANK
YOU

